

# Phase 4: Data Migration, Testing & Security

**Goal:** confirm the system is correct, secure, and reliable before deployment.

---

## Data Handling Review

- Only journey/passenger-count data stored — no payment or sensitive PII
- QR field always linked to exactly one ticket, never shared/reused
- Standard audit fields (created by/on) inherited for traceability

## Migration & Backup Approach

- Built directly on a fresh custom table — no legacy data migration required
- Table update sets exported and version-tagged after each working milestone, so any phase could be rolled back if a change broke automation

## Role-Based Access Control

| Role          | Access                        |
|---------------|-------------------------------|
| metro_user    | Create/view own bookings only |
| metro_support | View/manage all tickets       |
| admin         | Full configuration access     |

### ACLs applied at 3 levels:

- Table-level — controls table access overall
- Record-level — passengers see only their own tickets
- Field-level — hides internal fields (e.g. raw QR payload) from passengers

Server-side checks in Script Includes prevent business logic being bypassed client-side.

## Additional Security Measures

- Session handling relies on native ServiceNow authentication (no custom login)
- Script protections prevent direct field tampering via browser tools
- QR payload contains only ticket metadata, no credentials

## QA Test Plan & Execution

| Test Case                       | Expected Result                | Status |
|---------------------------------|--------------------------------|--------|
| Valid booking submission        | Ticket created                 | ✓ Pass |
| Same source/destination         | Blocked, validation error      | ✓ Pass |
| Fare calculation on known route | Correct fare applied           | ✓ Pass |
| New ticket submission           | Unique ticket number generated | ✓ Pass |
| Successful ticket creation      | QR generated & attached        | ✓ Pass |
| Successful ticket creation      | Confirmation email sent        | ✓ Pass |
| Cross-user record access        | Access denied                  | ✓ Pass |

**Process:** define input & expected result first → execute via portal (not just backend) → log pass/fail → trace failures to exact rule/script/ACL and fix at source.

## Performance Check

- Submitted multiple concurrent test bookings to confirm no ticket-number collisions or fare miscalculations under simultaneous inserts

## Data Integrity Check

- Fare/ticket-numbering rules apply identically via portal or direct admin insert
- QR never generated on a failed/incomplete insert
- Record-level ACLs hold under concurrent multi-user testing

## Bug Tracking

- Issues found during testing logged with: steps to reproduce, expected vs. actual result, owning component (rule/script/ACL)
- Fixed and re-tested before being marked closed

**Deliverable:** documented test log covering all Phase 1 scope items + a locked-down access model carried into deployment.